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Research on SSVEP Signal Classification Based on CNN-Mamba

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***Abstract*—Steady-State Visual Evoked Potential (SSVEP) signal classification is essential for Brain-Computer Interface (BCI) systems and rehabilitation medicine, yet accurate, low-latency classification of noisy electroencephalogram (EEG) signals in complex environments remains challenging. This paper presents a CNN-Mamba based SSVEP classification network, which integrates filter bank technology and Fast Fourier Transform (FFT) to convert time-domain signals into multi-subband frequency-domain features for enhanced frequency component perception. The Mamba architecture's Selective Scan mechanism is adopted for the backbone network, enabling efficient capture of global dependencies in long-sequence EEG signals with linear computational complexity. Combined with Convolutional Neural Network (CNN)'s strengths in local feature extraction, the model realizes multi-time-scale feature fusion and enhancement. Experimental tests on public SSVEP datasets demonstrate that the model achieves superior performance in classification accuracy and Information Transfer Rate (ITR).**

Keywords—Steady-State Visual Evoked Potential; Fourier Transform; CNN-Mamba; Spatiotemporal Convolutional Network; Filter Bank

I. INTRODUCTION

In recent years, deep learning has made remarkable progress in the field of electroencephalogram (EEG) signal processing, yet SSVEP classification in complex environments still faces prominent challenges. First, the time-frequency feature extraction capability is insufficient under low signal-to-noise ratio conditions, making it difficult to capture subtle frequency variations. Second, it is a tough task to balance the capture of global dependencies and inference efficiency in long-sequence modeling. To address these issues, this paper proposes a CNN-Mamba network. This model introduces filter banks and Fast Fourier Transform (FFT) to enhance frequency-domain perception, and leverages the Selective Scan mechanism of Mamba to resolve the efficiency and memory bottlenecks in long-sequence modeling[1]. Moreover, it combines CNN for local feature extraction to realize multi-scale feature fusion. Experimental results demonstrate that the proposed model achieves excellent performance in SSVEP classification tasks.

SSVEP signal classification has evolved from traditional Canonical Correlation Analysis (CCA) to end-to-end feature extraction based on Convolutional Neural Networks (CNNs). Recently, State Space Models (SSMs) have effectively addressed the bottleneck of Transformers in processing long sequences by virtue of their linear computational complexity. Among them, the S4 architecture[2] constructs a framework for capturing long-range dependencies through the HiPPO initialization strategy[3]. Mamba further incorporates the Selective Scan mechanism and time-varying parameters[4], which retains the high inference efficiency of RNNs while achieving Transformer-like parallel training capability, thus enhancing the dynamic modeling of non-stationary physiological signals.

Based on the above advances, this paper proposes the CNN-Mamba model. The model integrates filter banks and Fast Fourier Transform (FFT) to strengthen frequency-domain perception, leverages a Spatial-Temporal CNN to extract local features, and introduces a Mamba module to capture global dependencies. Experimental results show that the proposed method achieves excellent performance in SSVEP classification tasks.

II. MODEL INTRODUCTION

First, the CNN-Mamba model adopts a serial hybrid deep learning architecture to process EEG signals, as shown in Figure 1. In the front-end feature extraction part, local features are extracted step by step through the spatiotemporal convolution module: first, spatial convolution is used to learn the combination patterns of electrode signals at different positions, and then temporal convolution focuses on mining temporal features between sampling points. In the back-end sequence modeling part, the data is reshaped into an adaptive format through dimension permutation, and then the Mamba block is used to efficiently capture the global dependency features in long-duration data. Finally, the network calculates and predicts the probability of the signal belonging to each category through the MLP head.

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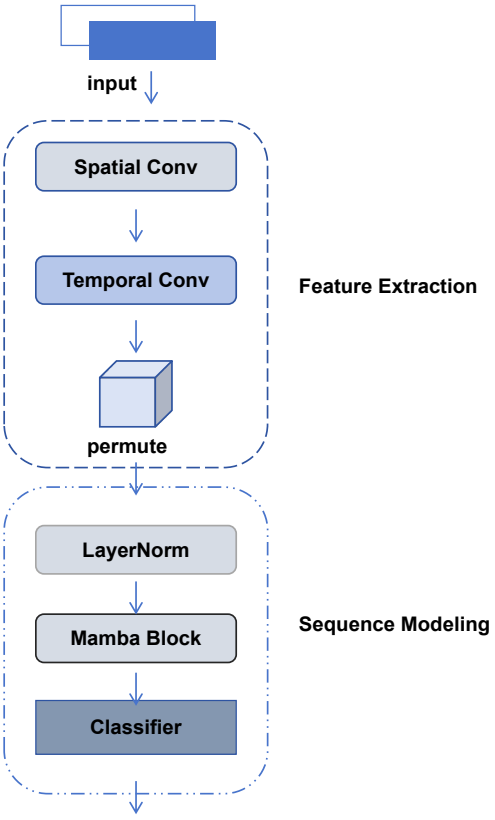


Figure 1. Structure diagram.

A. Feature Enhancement Module

The aggregation of spatial and temporal information is critical to the classification accuracy of SSVEP signals. To address the problems of low signal-to-noise ratio of single-channel EEG signals and the difficulty in capturing the collaborative activities of global brain regions, this paper proposes the following spatiotemporal feature extraction module.

1) Adaptive Spatial Convolution

To enhance the network's perception of the spatial distribution of EEG signals, this module introduces an adaptive spatial convolution mechanism. Drawing on the spatial filtering idea in EEGNet[5], this mechanism maps multi-channel raw EEG signals to a spatial feature subspace with strong discriminability.

Spatial convolution is a learnable spatial perception module that integrates the topological structure of the cerebral cortex into feature weights and representations, which can significantly improve classification performance. Since SSVEP signals are a collection of multi-channel time series, their effective features are completely determined by the synchronous firing activities of specific brain regions. If the spatial relationships between electrodes are not explicitly encoded, the network can only observe changes in a single time dimension but cannot capture the differences in global potential distribution induced by visual stimulation. We introduce spatial convolution, defined as the weighted aggregation across the channel dimension at time t, with the specific formula as follows:

$$Y_{spatial} = \sum_{c=1}^{C} (W_c \cdot X_{c,t} + b) \quad (1)$$

where $X_{c,t}$ denotes the input signal of the c-th electrode channel at time t, and W_c is the weight of the learnable spatial filter corresponding to this channel, which determines the contribution of the electrode position to feature extraction. In this way, the subsequent temporal convolution and Mamba module can better understand spatial features, thus achieving accurate classification of SSVEP signals.

2) Adaptive Temporal Convolution

To enhance the network's perception of the dynamic temporal changes of signals, this module introduces a temporal convolution mechanism. Drawing on the time-domain filtering idea in EEGNet and FBCNN[6], this mechanism maps the spatially filtered EEG signals to a high-dimensional feature space.

Temporal convolution is a learnable time-domain perception module that integrates the frequency fluctuations of EEG signals into feature representations, which can significantly improve classification performance. Since SSVEP signals are typical non-stationary time series, their core semantics are completely determined by changes in frequency and phase. If temporal dependencies are not explicitly extracted, the network can only observe the instantaneous voltage amplitude but cannot capture the periodic rhythmic information induced by visual stimulation. We introduce temporal convolution, defined as a one-dimensional convolution operation on the spatial feature sequence, with the specific formula as follows:

$$Y_t = ReLU(\sum_{k=1}^{K} W_k \cdot X_{t:t+L} + b) \quad (2)$$

where W_k denotes the weight of the k-th temporal convolution kernel, acting as a learnable band-pass filter responsible for extracting time-domain textures related to the frequency of visual stimulation from the signal; $X_{t:t+L}$ denotes the input of a local time window with length L. In this way, the subsequent Mamba module can better understand long-range temporal dependencies, thus achieving accurate classification of SSVEP signals.

B. Mamba Module

Sequence modeling serves as a critical component in SSVEP signal classification networks, responsible for capturing long-range dependencies and extracting global features from temporal signals. With the increase in sampling rate and signal length, although Transformer can capture global dependencies via self-attention mechanisms, its quadratic computational complexity leads to a significant rise in GPU memory consumption and inference latency, which fails to meet the stringent real-time requirements of brain-computer interface systems.

In this paper we introduce the Mamba module based on the Selective Scan mechanism, aiming to efficiently filter key temporal information while maintaining linear computational

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complexity. We replace the traditional attention mechanism with the Mamba architecture, and balance inference efficiency and long-range dependency capture capability by optimizing the state space model.

State-space sequence models (SSMs), particularly the S4 architecture, are proposed with the core idea of discretizing differential equations of continuous systems to process sequential data, enabling the network to handle long sequences more effectively. However, traditional linear time-invariant (LTI) state-space models typically employ fixed parameter matrices (A, B, C) to process inputs at all time steps, which prevents the model from dynamically adjusting state weights based on the current input content. Although this approach reduces computational complexity to linear order, it overlooks the time-varying contextual differences of signals. As a result, transient features associated with visual stimulation in SSVEP signals tend to be smoothed out, failing to adequately reflect the significance of signals within different time windows.

This paper optimizes the structure of the state space model and introduces the unique Selective Scan mechanism of Mamba. As shown in Figure 2, to balance computational efficiency and content perception capability, we incorporate time-varying parameters into the state equation. The feature sequence after dimension permutation is first passed through a linear projection layer to make the feature sequences a function of the current input X_t. Then, the main branch is processed by a short convolution and an activation function, and then the hardware-aware SSM parallel scanning algorithm is used to perform recursive state update efficiently on the GPU. Finally, we combine the updated hidden state with the dynamic output matrix of the gating branch to generate a new feature representation with a global receptive field. This enables the network to not only have linear time complexity in sequence modeling but also selectively compress or retain memory according to the input content. Finally, we embed this Mamba module into the backbone of the SSVEP classification network.

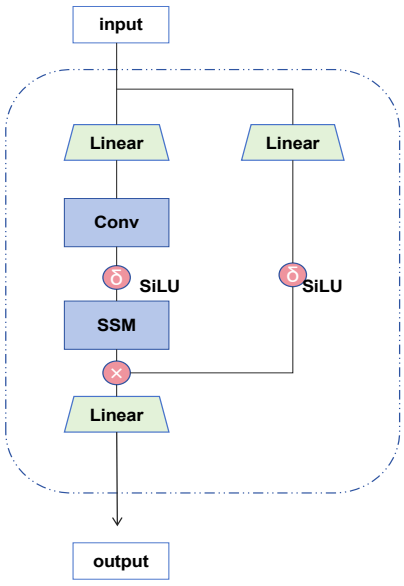


Figure 2. Mamba module structure

III. Experiment

A. Experiment Preparation

The experiments in this paper are conducted on a system equipped with an NVIDIA 3060 (8G) GPU, with Ubuntu 24.04 as the operating system, and CUDA 11.8, PyTorch 2.2.2 and Python 3.8 as the development environment. The network training parameters are set as follows: the batch size is 64, the Adam optimizer is adopted, and the learning rate is halved when the loss of the validation set does not decrease for 5 consecutive epochs. The experimental data are from the public BETA and Benchmark datasets[7], and five-fold cross-validation is adopted.

B. Evaluation Metrics

This paper selects two evaluation metrics commonly used in SSVEP signal classification tasks, including classification accuracy (Acc) and Information Transfer Rate (ITR). ITR is a key metric for evaluating the comprehensive performance of BCI systems, especially SSVEP-BCI systems. It not only measures the consistency between the model's predicted categories and the real stimulation labels (i.e., accuracy) but also comprehensively considers the number of classification targets and the time required for single target recognition, reflecting the communication efficiency of the system per unit time. The calculation formula is as follows:

$$ITR = \frac{60}{T} \times [log_2 N + log_2 P + (1-P) \times log_2 \frac{1-P}{N-1}] \quad (3)$$

where N is the number of classification targets, P is the classification accuracy, and T is the time required for single target selection.

To evaluate the performance of CNN-Mamba, we use two public datasets (BETA, Benchmark) for verification. These datasets cover different subject groups, stimulation frequency ranges (8Hz-15.8Hz) and phase settings. The average accuracy and information transfer rate of 5-fold cross-validation are regarded as performance metrics, and experimental results show that compared with the baseline models, our network model exhibits superior robustness at different time scales.

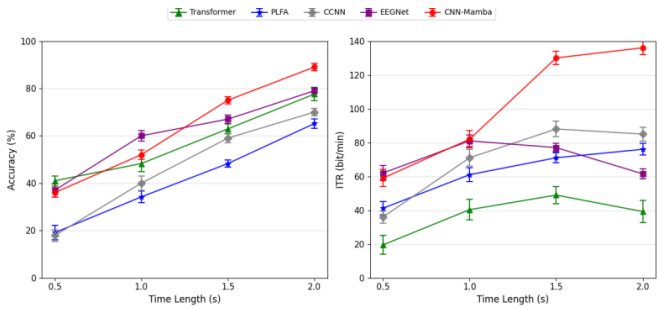


Figure 3. Classification results

We compare the Acc and ITR metrics with traditional baseline models such as Transformer and deep learning models including EEGNet[8], CCNN[9] and PLFA[10]. The specific results are shown in Figure 3. Compared with the classic convolutional network EEGNet, CNN-Mamba has a significant improvement

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in both metrics. Taking the BETA dataset as an example (2.0s time window), the Acc of CNN-Mamba reaches 89.44% and the ITR reaches 136.09 bit/min.

IV. Conclusion

To solve the common problems of low signal-to-noise ratio and insufficient time-frequency feature extraction in SSVEP signal classification, this paper proposes the CNN-Mamba network. By combining the spatial convolution and temporal convolution modules, the model not only focuses on the spatial topological combination of EEG signals but also has a better grasp of the local texture features in the time dimension. In this way, the network can more fully retain key frequency-domain information and temporal changes in the feature mapping process, thereby strengthening the model's ability to identify weak frequency stimulation signals and short-time-window data.

In the sequence modeling stage, the network uses the unique Selective Scan mechanism of the Mamba architecture to filter the serialized deep features. Utilizing its hardware-aware parallel algorithm and dynamic state adjustment capability, the model actively focuses on task-related signals in long sequences and weights key temporal dependencies. Through dynamic parameter adjustment, it alleviates the forgetting phenomenon of traditional recurrent neural networks in long-sequence processing and suppresses the impact of environmental noise and biological artifacts on prediction. Meanwhile, this architecture takes into account the high efficiency of RNNs, making the extracted global features more stable and the overall Information Transfer Rate (ITR) of the system improved. Finally, excellent classification results are achieved on public SSVEP datasets such as Benchmark and BETA.

In future research, we will conduct a detailed complexity analysis including FLOPs, GPU memory consumption and inference time, and compare the proposed method with Transformer baselines to evaluate its efficiency on embedded BCI systems.For the Mamba selection mechanism, we will further visualize the selection masks and state dynamics to reveal its attention distribution on target stimulation frequencies and non-harmonic noise.

References

[1] Gu A, Dao T. Mamba: Linear-time sequence modeling with selective state spaces[J]. arXiv preprint arXiv:2312.00752, 2023.

[2] A. Salam, R. Mahmud, T. Islam, S. Mukta and S. Shatabda, "A Comprehensive Survey on Mamba: Architectures, Challenges, and Opportunities," in Computer, vol. 58, no. 8, pp. 64-76, Aug. 2025, doi: 10.1109/MC.2025.3571322.

[3] Gu A, Goel K, Ré C. Efficiently modeling long sequences with structured state spaces[J]. arXiv preprint arXiv:2111.00396, 2021.

[4] Gu A, Johnson I, Goel K, et al. Combining recurrent, convolutional, and continuous-time models with linear state space layers[J]. Advances in neural information processing systems, 2021, 34: 572-585.

[5] Rao, Yongkang, Le Zhang, Ruijun Jing, Jiabing Huo, Kunxian Yan, Jian He, Xiaojuan Hou, Jiliang Mu, Wenping Geng, Haoran Cui, Zeyu Hao, Xiang Zan, Jiuhong Ma and Xiu-jian Chou. “An optimized EEGNet decoder for decoding motor image of four class fingers flexion.” Brain research (2024): 149085.

[6] Wen X, Jia S, Han D, Dong Y, Gao C, Cao R, Hao Y, Guo Y, Cao R. Filter banks guided correlational convolutional neural network for SSVEPs based BCI classification. J Neural Eng. 2024 Oct 4;21(5). doi: 10.1088/1741-2552/ad7f89.

[7] M. Young, The Technical Writer’s Handbook. Mill Valley, CA: University Science, 1989.

[8] Lawhern V J, Solon A J, Waytowich N R, et al. EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces [J]. Journal of neural engineering, 2018, 15(5): 056013.

[9] Ravi A, Beni N H, Manuel J, et al. Comparing user-dependent and user-independent training of CNN for SSVEP BCI[J]. Journal of neural engineering, 2020, 17(2): 026028.

[10] LIN Y, ZANG B, GUO R, et al. A deep learning method for SSVEP classification based on phase and frequency characteristics[J]. Journal of Electronics and Information Technology, 2022, 44(2): 446-454.

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